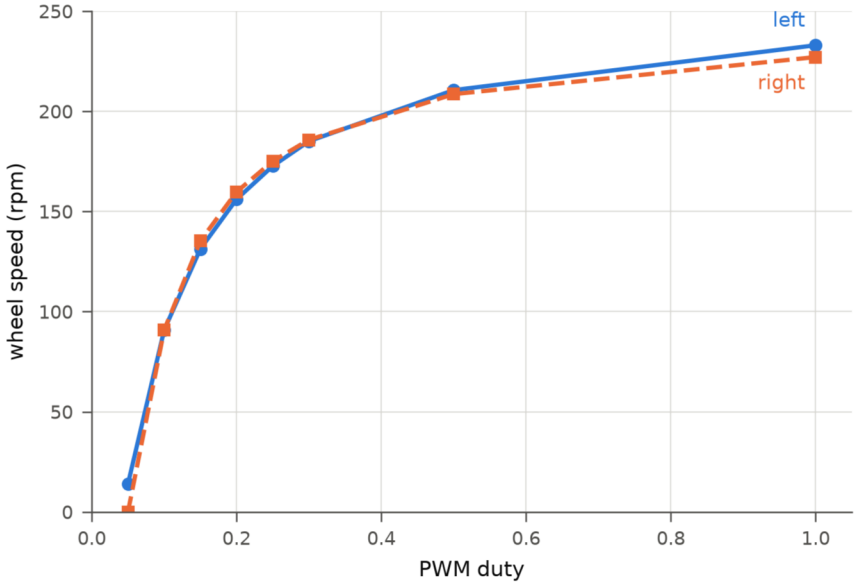


Figure 7-1. Wheel speed against PWM duty, both wheels, rover on blocks. Measured 13 Oct 2026, fullspeedtest.py rev 2a841e1, data in plant.csv. Steep below duty 0.2, nearly flat above 0.3: going from 0.3 to 1.0 adds only 25 percent more speed. The right wheel does not turn at duty 0.05, placing the deadband there.

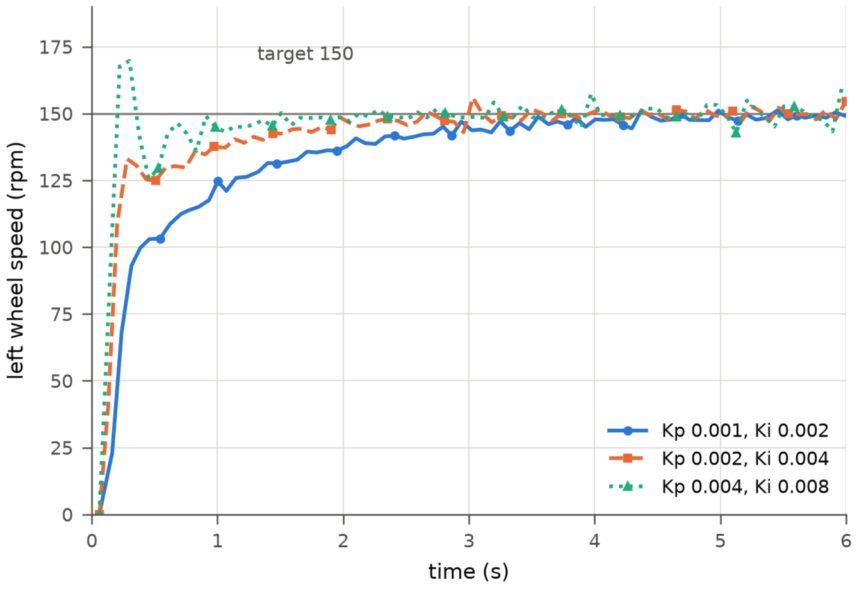


Figure 7-2. Step response of the left wheel to a 150 rpm target at three gain settings, rover on blocks. Measured 13 Oct 2026, pidtest.py rev 2a841e1, data in pid_150_*.csv. Kp 0.001 reaches 90 percent in 1.71 s with no overshoot, Kp 0.002 in 0.82 s, Kp 0.004 overshoots to 170 rpm at t 0.30 s then rings. Chosen gains are half the oscillation point.